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Submission date: 02-Apr-2026 08:38AM (UTC+0300)

Submission ID: 2920488502

File name: Manuscript_4_.docx (37.54K)

Word count: 2230

Character count: 14568

Type of Manuscript:
Research

Comparative Analysis of Lipid Profiles Between ²Controlled and Uncontrolled Type 2 Diabetes Mellitus Patients at Tarakan Regional General Hospital (2022–2023)

ABSTRACT⁷

Background: ⁷Type 2 diabetes mellitus (T2DM) contributes substantially ²to cardiovascular complications, largely through disturbances in lipid metabolism associated with inadequate glycemic ¹⁵control. However, evidence from hospital-based populations in Indonesia remains limited. This study aimed to evaluate differences ¹³in lipid profiles between patients with controlled and uncontrolled T2DM.

Methods: A cross-sectional analytic study was conducted using medical records of 120 patients with T2DM. Patients were classified into ²controlled (HbA1c <7%) and uncontrolled (HbA1c ¹⁰≥7%) groups. Lipid parameters assessed included total cholesterol, LDL-C, HDL-C, and triglycerides. Statistical analysis was performed using appropriate parametric and non-parametric tests.

Results: Patients with ³uncontrolled T2DM exhibited significantly higher total cholesterol, LDL-C, and triglyceride levels compared with controlled patients. ¹¹No statistically significant difference was observed in HDL-C levels. Dyslipidemia was more prevalent in patients with poor glycemic control.

Conclusion: Uncontrolled T2DM is associated with unfavorable lipid alterations, emphasizing the importance of maintaining optimal glycemic control to reduce cardiovascular risk.

Keywords: diabetes mellitus, HDL, total cholesterol, LDL, triglycerides

9 INTRODUCTION

The increasing prevalence of type 2 diabetes mellitus (T2DM) has become a major concern due to its contribution to long-term vascular complications, particularly cardiovascular disease. Indonesia is among the countries with a substantial number of diabetes cases, with approximately 19.5 million individuals affected in 2021.¹ National surveys have also demonstrated a steady rise in prevalence from 6.9% in 2013 to 8.5% in 2018, indicating a growing public health burden.²

Metabolic dysregulation in T2DM²⁹ is closely linked to insulin resistance and chronic hyperglycemia, which³ interfere with normal lipid homeostasis. These disturbances often result in an unfavorable lipid profile characterized by elevated triglycerides and low-density lipoprotein cholesterol (LDL-C), along with reduced high-density lipoprotein cholesterol (HDL-C).³ Such alterations contribute to the progression of atherosclerosis and increase the likelihood of cardiovascular complications.^{3–5} Prolonged exposure to hyperglycemia may also trigger oxidative stress and endothelial dysfunction, further accelerating vascular damage.^{3–5}

Glycemic status, commonly assessed using glycated hemoglobin (HbA1c), is an important determinant of these metabolic changes. Patients with poor glycemic control tend to exhibit more pronounced lipid abnormalities compared with those whose glycemic levels are adequately controlled.^{4,5} Despite this well-established relationship, evidence derived from hospital-based populations in Indonesia remains scarce.

Based on this background,²⁸ the present study was designed to evaluate differences in lipid profiles between controlled and uncontrolled T2DM patients and to explore their clinical relevance in relation to cardiovascular risk.

14 METHODS

This analytic cross-sectional⁸ study was conducted at a tertiary care hospital in Jakarta, Indonesia, in October 2024. Ethical approval was obtained from the Institutional Ethics Committee (No. 1811/SLKE/TM/UKK W/FKIK/KEPK/TX/2024).

A total of 120 patients with type 2 diabetes mellitus¹⁶ (T2DM) were included using consecutive sampling from medical records. T2DM was defined based on documented glycated hemoglobin (HbA1c) $\geq 6.5\%$ or fasting plasma glucose ≥ 126 mg/dL. Patients with incomplete lipid profile data, those receiving lipid-lowering therapy (including statins, fibrates, niacin, or ezetimibe),⁶ and those with chronic kidney disease, hyperthyroidism, or liver cirrhosis were excluded.

Patients were categorized into controlled (HbA1c $< 7\%$) and uncontrolled (HbA1c $\geq 7\%$) groups. The independent variable⁷ was glycemic control status, while the dependent variables included lipid parameters: triglycerides, low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C), and total cholesterol. Lipid profile measurements were obtained from medical records and were assessed within a clinically relevant timeframe relative to glycemic evaluation.²³

Data were analyzed using⁴ SPSS version 24. Normality of data distribution was assessed prior to analysis. Independent t-test was used for normally distributed variables, while the Mann-Whitney U test was applied for non-normally distributed data. A p-value < 0.05 was considered statistically significant.

12 RESULTS

A total of 120 patients with type 2 diabetes mellitus (T2DM) were included in this study. The variables presented in Table 1 include demographic characteristics (age and sex) and anthropometric measurements, specifically body mass index (BMI). Most participants were in the pre-elderly age group (45–59 years), comprising 62 individuals (52.7%). Females represented the majority of the cohort (n = 73; 60.8%). Based on body mass index (BMI), Obesity Class I (25–29.9 kg/m²) was the most prevalent category (n = 49; 40.8%). Furthermore, uncontrolled diabetes was observed in 77 patients (64.2%), indicating that poor glycemic control predominated in this population.

18 **Table 1.** Distribution of Study Participants According to Demographic and Clinical Characteristics

Characteristics	Frequency (%)	Cholesterol Level			
		Total Cholesterol	Triglycerides	LDL	HDL
		(mg/dL)			
Age(years)					
Adult(19-44)	10 (8,3)	182,5 (105-324)	130 (68-193)	119 (60-225)	42 (12-60)
Pre-Elderly(45-59)	62 (52,7)	212 (108-323)	160 (57-828)	121 (46-211)	46 (23-75)
Elderly (≥60)	48 (40,0)	191 (95-299)	131 (49-366)	106 (19-209)	45 (23-79)
Sex					
Male	47 (39,2)	177 (95-315)	142 (49-828)	110 (19-209)	44 (12-73)
Female	73 (60,8)	216 (123-343)	143 (57-617)	129 (45-225)	50 (23-79)
BMI (kg/m²)					
Underweight (<18,5)	7 (5,8)	205 (145-218)	195 (75-285)	121 (80-144)	43 (26-60)
Normal (18,5-22,9)	14 (11,7)	182 (135-235)	116 (57-213)	101 (45-159)	43 (27-79)
Overweight(23-24,9)	24 (20,0)	186 (108-343)	129 (49-480)	108 (46-209)	45 (23-75)
Obesity Class I (25-29,9)	49 (40,8)	212 (95-324)	149 (84-828)	124 (19-225)	47 (12-75)
Obesity Class II (≥30)	26 (21,7)	194 (123-264)	150 (66-385)	117 (75-211)	47 (23-74)
Glycemic Control Status					
Controlled (< 7%)	43 (35,8)	179,23 ±43,24	128 (49-385)	109,70 ±38,66	45,86 ±13,66
Uncontrolled (≥ 7%)	77 (64,2)	210,86 ±51,64	156 (57-828)	122,62 ±37,72	47,29 ±13,68

Analysis of lipid parameters across age groups showed that the pre-elderly group exhibited the highest median concentrations of total cholesterol [212 (108–323) mg/dL], triglycerides [160 (57–828) mg/dL], LDL-C [121 (46–211) mg/dL], and HDL-C [46 (23–75) mg/dL], suggesting a clustering of lipid abnormalities within this age category. When stratified by sex, female patients demonstrated higher median lipid levels compared with males. Median total cholesterol reached 216 (123–343) mg/dL and LDL-C 129 (45–225) mg/dL among females, indicating a less favorable lipid profile in this subgroup.

Across BMI categories, patients classified as Obesity Class I had the highest median total cholesterol and LDL-C levels, measuring 212 (95–324) mg/dL and 124 (19–225) mg/dL, respectively. In contrast, the highest median triglyceride concentration was observed in the underweight group at 195 (75–285) mg/dL, demonstrating variability in lipid distribution across nutritional status. Stratification according to glycemic control revealed that patients with uncontrolled diabetes consistently exhibited higher lipid levels, including total cholesterol of $210 \pm 51,64$ mg/dL, triglycerides of 156 (57–838) mg/dL, and LDL-C of $122,62 \pm 37,72$ mg/dL.

Table 2. Distribution of Dyslipidemia Among Patients with Type 2 Diabetes Mellitus

Dyslipidemia	Glycemic Control Status	
	Controlled (n=43)	Uncontrolled (n=77)
	n (%)	
Dyslipidemia	4 (3,3)	12 (10,0)

As shown in Table 2, dyslipidemia was more prevalent among patients with uncontrolled diabetes, accounting for 12 individuals (10.0%) of the study population.

Table 3. Comparison of Lipid Parameters According to Glycemic Control Status

Lipid Profile	Glycemic Control Status		Nilai p
	Controlled (n=43)	Uncontrolled (n=77)	
LDL	109,70 $\pm$ 38,66	122,62 $\pm$ 37,72	0,047 ^a
HDL	45,86 $\pm$ 13,66	47,29 $\pm$ 13,68	0,585 ^a
Kolesterol Total	179,23 $\pm$ 43,24	210,86 $\pm$ 51,64	0,001 ^a
Trigliserida	131,66 (49-285)	156,00 (57-828)	0,013 ^b

^aUji T test independent, ^bUji mann whitney, significant if $p < 0,05$

The results demonstrated significant differences in total cholesterol ($p = 0.001$), LDL-C ($p = 0.047$), and triglyceride levels ($p = 0.013$) between patients with controlled and uncontrolled type 2 diabetes mellitus. Overall, patients with uncontrolled diabetes exhibited higher lipid levels compared with those with controlled diabetes. The mean LDL-C level was 122.62 ± 37.72 mg/dL, and the mean total cholesterol level was 210.86 ± 51.64 mg/dL. Meanwhile, the median triglyceride level was 156.00 (57–828) mg/dL.

DISCUSSION

This study showed that the majority of patients with T2DM were within the 45–59 year age group, which aligns with previous reports indicating that metabolic disorders are more frequently observed in individuals above 45 years.^{9,10} Age-related physiological decline, together with lifestyle-related factors, may contribute to impaired regulation of both glucose and lipid metabolism.^{10,11}

Female patients represented a larger proportion of the study population and tended to exhibit higher lipid levels compared with males. Similar findings have been reported in earlier studies, suggesting that women with T2DM may be more prone to dyslipidemia.^{7,12} Hormonal changes, particularly decreased estrogen levels, may contribute to increased visceral fat accumulation and insulin resistance, thereby influencing lipid metabolism.¹¹

In terms of nutritional status, most patients were classified as Obese²⁴ Class I, and this group showed higher total cholesterol and LDL-C levels. This observation is consistent with previous evidence demonstrating a strong association between obesity and lipid abnormalities in T2DM.^{13,14} Excess adipose tissue can increase the release of free fatty acids and inflammatory mediators, which interfere with insulin signaling and enhance hepatic lipoprotein production.^{13,15}

Notably, lipid abnormalities were also identified among underweight individuals, particularly elevated triglyceride levels. Although this finding is less commonly reported, it suggests that dyslipidemia in T2DM is not exclusively linked to obesity and may also be influenced by broader metabolic disturbances.^{14,16} This highlights the complex nature of lipid metabolism in diabetic conditions.

The main finding of this study indicates that patients with uncontrolled T2DM had significantly higher levels of total cholesterol, LDL-C, and triglycerides compared with those with controlled glycemic status. These findings are consistent with previous studies reporting that poor glycemic control is associated with a more atherogenic lipid profile.^{4,5,15} Elevated LDL-C levels in uncontrolled diabetes may be related to increased hepatic production of very low-density lipoprotein (VLDL) and subsequent formation of atherogenic lipoproteins.^{18,19}

²² In contrast, HDL-C levels did not differ significantly between the two groups. Although previous studies generally report reduced HDL-C levels in T2DM, this discrepancy may be influenced by differences in patient characteristics, treatment adherence, or sample size.¹⁹

Higher triglyceride levels in patients with poor glycemic control further support the role of insulin resistance in lipid metabolism disturbances. Increased lipolysis and the release of free fatty acids contribute to enhanced hepatic triglyceride synthesis, thereby promoting dyslipidemia.¹⁹

Overall, the findings emphasize the importance of achieving adequate glycemic control in patients with T2DM. Poorly controlled diabetes is associated with a more atherogenic lipid profile, which may increase cardiovascular risk. Therefore, integrated management strategies that include both glycemic control and lipid monitoring are essential to improve clinical outcomes.²⁰

CONCLUSION

⁷ This study demonstrates that glycemic status plays a critical role in shaping lipid abnormalities in patients with type 2 diabetes mellitus. Uncontrolled diabetes was consistently associated with significantly higher levels of total cholesterol, LDL-C, and triglycerides, reflecting a more atherogenic lipid profile and a greater potential for cardiovascular complications. Notably, lipid disturbances were observed across different BMI categories, including underweight individuals, indicating that dyslipidemia in T2DM is not solely attributable to obesity but arises from complex metabolic interactions involving insulin resistance, glycemic control, and lipid metabolism. These findings underscore the importance of comprehensive metabolic assessment and strict glycemic management, alongside routine lipid monitoring, to reduce cardiometabolic risk and improve long-term outcomes in patients with T2DM.

Conflict of Interest

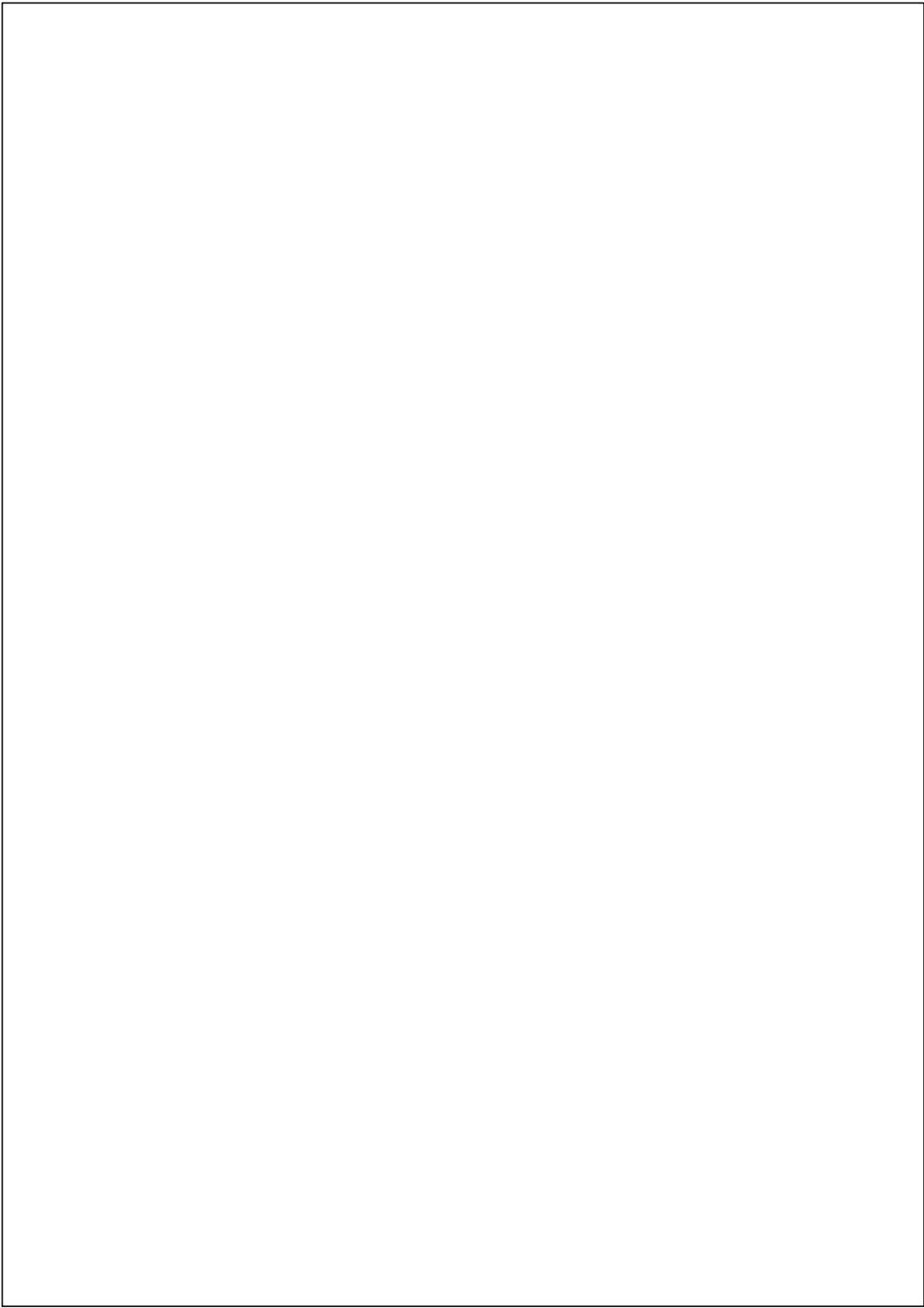
The authors declare that there are no conflicts of interest regarding the publication of this study.

Funding

This research received no external funding.

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